Speaker Script — Presenter 1

Speaker Script — Presenter 1 (David, Group 23)

Scope: Intro + Part 1 (Problem, Data & Pre-processing) + Part 2 (Three Approaches) + Part 3 (Approach 1 — Classical ML). Hands off to **Hoa** (Presenter 2), who takes Approach 2.

Target runtime: 4 minutes. 6 slides, each showing a single plot at readable size. The goal is to show we understood *why* we did what we did, not to read numbers off the slide. Word counts tuned for ~165 wpm delivery with ~20 s of overall buffer.

Pacing guide:

Slide	Time	Budget
1. Title + Research Question	0:00–0:45	45s
2. Dataset & Pre-processing — Decisions With a Reason	0:45–1:45	60s
3. Three-Approach Attribution Design	1:45–2:45	60s
4a. A1 — Architecture beats depth	2:45–3:15	30s
4b. A1 — Resolution dominates	3:15–3:40	25s
4c. A1 — Classifier barely matters + handoff	3:40–4:00	20s

Slide 1 — Title + Research Question (~45s, ~115 words)

Good [morning / afternoon] — we're Group 23: David, Hoa, and Khoa.

Our task is image classification on Mini-ImageNet — 100 classes.

But the question we really want to answer isn't "can we classify these images?" It's "where does our accuracy come from?" Is it from pretraining? From fine-tuning? Or from training the network from scratch on our data?

To answer that, we run three approaches side by side: frozen pretrained features, fine-tuning, and training from scratch. Each one isolates a different source, so we can tell which part matters.

I'll cover the problem, the data, and Approach 1. Hoa takes Approach 2. Khoa finishes with Approach 3 and the cross-approach analysis.

Slide 2 — Dataset & Pre-processing: Decisions With a Reason (~47s, ~130 words)

Mini-ImageNet has 100 classes, split fifty / ten / five thousand for train, val, and test. The classes are perfectly balanced — you can see that in the plot — so imbalance won't skew our later results.

We made three pre-processing choices, each with a reason.

First, two resolutions. 224 matches what the backbones were pretrained on — our best case. 32 is a deliberate stress test. It forces models to work outside what they were built for, which is where architecture choices — especially the first layer — really start to matter.

Second, our own normalisation stats, not ImageNet defaults. Shrinking to 32 pixels shifts the pixel values, so ImageNet's numbers would be slightly off for us.

Third, different augmentation per approach. Approach 1 caches features once, so no augmentation during training. Approaches 2 and 3 train end-to-end with random crops and flips.

Slide 3 — Three-Approach Attribution Design (~55s, ~150 words)

Three approaches, each answering a different question.

Approach 1 — frozen features. A pretrained backbone with frozen weights plus a simple linear classifier. Tells us how good pretrained features are on their own.

Approach 2 — fine-tuning. Start from pretrained weights, let some layers update. Our proposed improvement. Tells us how much adapting to Mini-ImageNet adds on top.

Approach 3 — from scratch. Random starting weights, no ImageNet head start. Tells us how much signal is in Mini-ImageNet alone.

Why run all three? A single accuracy number mixes pretraining, adaptation, and in-domain learning into one lump. Running three separates them.

Why pretrained models in the first place? As a well-understood baseline to build on. A1 uses them frozen — our starting point. A2 fine-tunes them — our proposed improvement. A3 trains the same architectures from scratch. The pretrained backbone is the common anchor our approaches measure against.

Slide 4a — A1: Architecture Beats Depth at 32×32 (~31s, ~85 words)

At 32 pixels, architecture matters far more than depth. ResNet-18, 34, and 50 all sit around 33% despite roughly 2× the parameters — depth doesn't help. But ConvNeXt-Tiny jumps up by ten-plus points.

The reason is the stem — the first layer. ConvNeXt's 4-by-4 patchify shrinks the image in one clean step and leaves enough structure for the later layers to work with. ResNet's bigger conv plus max-pool squeezes a 32-pixel input down too fast, and the deeper ResNets inherit the same bad stem.

Slide 4b — A1: Resolution Dominates Any Architecture Gap (~24s, ~65 words)

Resolution is the biggest single factor — bigger than any architecture gap. Both backbones lose about 50 points going from 224 to 32.

The t-SNE plot shows why. At 224, each class forms a tight, separable cluster. At 32, everything collapses together. Same backbone, same features — but at 32 pixels there's just no spatial extent left for the model to tell classes apart.

Slide 4c — A1: Classifier Barely Matters (~17s) + Handoff (~7s)

The classifier barely matters. SVM and logistic regression come out comparable for every backbone. That's a validation: we set out to measure feature quality, and swapping the classifier doesn't change the ranking.

We carry forward **ConvNeXt-Tiny** and **ResNet-18** into Approaches 2 and 3.

Over to Hoa — can fine-tuning close the ConvNeXt-vs-ResNet gap?

Likely Q&A — prep notes

- **"Why Mini-ImageNet instead of full ImageNet-1K?"** Two reasons — our compute budget, and the small-data setting (500 images per class) is exactly where the pretraining question is most interesting.
- **"Isn't using pretrained models cheating, since the classes overlap with ImageNet-1K?"** We're using them as a well-understood baseline to build on, not to claim transfer to new data. Approach 2 fine-tunes those same backbones, and Approach 3 trains the same architectures from scratch — so the pretrained model is the common anchor, and our contributions are what we add on top.
- **"Why compute your own mean and standard deviation instead of using ImageNet defaults?"** The 32-pixel resize shifts the pixel distribution a bit. Using our own stats gives cleaner normalisation for Approach 3. Fair point the other way: Approach 2 might slightly prefer ImageNet defaults, because pretrained BatchNorm is calibrated to them — but the effect is under 1 point.
- **"Why is ConvNeXt so much better at 32×32?"** The stem. ConvNeXt's 4-by-4 patch convolution keeps an 8-by-8 feature map, which later layers can work with. ResNet's stem collapses small input too fast. We haven't run a direct ablation, but this explanation fits the data.
- **"Why didn't you include ViT or Swin?"** We picked one representative from each architecture family. ConvNeXt stands in for the modern post-ViT CNN design.
- **"Does the 50-point drop at 32 pixels generalise?"** We've only shown it on two backbones on Mini-ImageNet — so we don't claim it as a general rule. But the explanation (small input collapsing in the stem) would predict something similar for other architectures with the same kind of stem.
- **"ResNet-50 uses V2 weights and the others use V1 — is that a problem?"** Yes, it's a real confound. V2 is about 4 points stronger on ImageNet-1K. To compare ResNet depth cleanly we'd want to pin all of them to V1.

- **“Why not tune hyperparameters?”** Compute budget, and our goal was to compare backbones, not optimise the classifier. Using sklearn defaults keeps the comparison simple and fair.
- **“Why two classifiers?”** To check that what we’re measuring is feature quality, not the classifier’s inductive bias. SVM and logistic regression land within 1 point for every backbone, which confirms that reading.